Student ID: 6888143 Name: Qimin Zhu Section: 1

ITCS216: Fundamentals of Database Systems (1/2026)

Faculty of Information and Communication Technology

In-class Activity #2: **Business Requirement Analysis**

Objectives:

- Understand the differences between data and information.
- Identify the process in the selected business domain.
- Identifying the key data in the given process in the business domain.

Instructions:

1. Open the online bookstore websites: “Kinokuniya” bookstore (https://thailand.kinokuniya.com/)
2. Answer the following questions and submit them by the end of the class.

Submission:

Submit your answer via MyCourses by the end of the class.

- If your file is already digitalized, rename the file to ITCS216-InclassActivity-Lect02-studentID.pdf
- If it is paper-based, upload the photo of your answer in MyCourses; also named the file to ITCS216-InclassActivity-Lect02-studentID

Use the website, https://thailand.kinokuniya.com/, to answer the following questions.

1. What are the main business processes that the business (Kinokuniya) offers to their customer?

   List at least 3 processes.

   Example:

Process:	Book Purchasing
Explanation:	The customers can purchase the book online. Customers review the items in their shopping cart, confirm quantities, and proceed to the checkout process. They may have the option to apply any applicable discounts or promotional codes at this stage. They must specify the delivery option, whether it’s to their address or for pick-up at the store. They are required to have the account prior to purchasing.

ANS

1. Review shopping cart
2. Check out
3. Account register

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2. For **each process**, list the *Data Object* and its *Data Properties* that are involved with the example/evidence if any. You must list **at least 1 data object** per process.

Example:

Process: Book Purchasing	
Data involve the process include:	
Book	title, price, author, images, publisher, item type, ISBN
Membership Account	first name, last name, Kinokuniya Privilege Card Number, expiry date
Billing address	house number, road, sub district, district, city, country, postal code

Process: Review shopping cart	
Data involve the process include:	
Item	title, price, author, images, quantities, item genre, ISBN, Publisher

Process: Check Out	
Data involve the process include:	
shipping address	first name, last name, address, city, country/region, state/province, Postal code, phone
billing address	first name, last name, address, city, country/region, state/province, Postal code, phone
Payment option	Payment type, payment identity, password

Process: Account management	
Data involve the process include:	
account	email, register date, name, age
My order	orderID, item, status
subscription	subscription plan, subscription fee, subscription duration

3. For **each process**, identify 3 business rules that are covered both Forward and Backward sentences, as well, identify the final conclusion for each rule.

Example:

Process: Book Purchasing	
Rule No. ***n***	
Forward sentence	A customer can buy many books. (1:M)
Backward sentence	A book can be bought by only one customer. (1:1)
Final conclusion	A customer can buy many books. (1:M)

Process: Review shopping cart	
Rule No.: 1 A shopping cart can cantain many books	
Forward sentence	A shopping cart can cantian many books. (1:M)
Backward sentence	A book can be added to many shopping cart. (1:1)
Final conclusion	A shopping cart can cantian many books. (1:M)

Process: Check Out	
Rule No.: 2 A delivery option can be selected by many customers.	
Forward sentence	A customer can only choose one delivery option.(1:1)
Backward sentence	A delivery option can be selected by many customers.(1:M)
Final conclusion	A delivery option can be selected by many customers.(1:M)

Process: Account management	
Rule No.: 3 A customer can have multiple account.	
Forward sentence	A customer can have multiple account.(1:M)
Backward sentence	An account can be owned by only one user.(1:1)
Final conclusion	A customer can have multiple account.(1:M)

Process: Review shopping cart	
Rule No.: 4 A book can be added to shopping cart multiple times .	
Forward sentence	A shipping cart can have multiple same book.(1:M)
Backward sentence	A book can be added to shopping cart multiple times .(1:M)
Final conclusion	A book can be added to shopping cart multiple times .(M:M)

Process: Check Out	
Rule No.: 5 A shipping address can be setted to many order.	
Forward sentence	An order can only have one shipping address.(1:1)
Backward sentence	A shipping address can be setted to many order. (1:M)
Final conclusion	A shipping address can be setted to many order. (1:M)

Process: Check Out	
Rule No.: 6 An shipping address as billing address.	
Forward sentence	A billing address can same as the shipping address.(1:1)
Backward sentence	A shipping address can be used for billing address.(1:1)
Final conclusion	A billing address can same as the shipping address.(1:1)

Process: Check Out	
Rule No.: 7 A payment method can be used for many payments.	
Forward sentence	A payment can only uses one payment method.(1:1)
Backward sentence	A payment method can be used for many payments.(1:M)
Final conclusion	A payment method can be used for many payments.(1:M)

Process: Account management	
Rule No.: 8 An account can have multiple orders.	
Forward sentence	An account can have multiple orders.(1:M)
Backward sentence	An order can be owned by one account.(1:1)
Final conclusion	An account can have multiple orders.(1:M)

Process: Account management	
Rule No.: 9 A subscription can be owned by many accounts.(1:M)	
Forward sentence	An account can only have one membership subscription.(1:1)
Backward sentence	A subscription can be owned by many accounts.(1:M)
Final conclusion	A subscription can be owned by many accounts.(1:M)